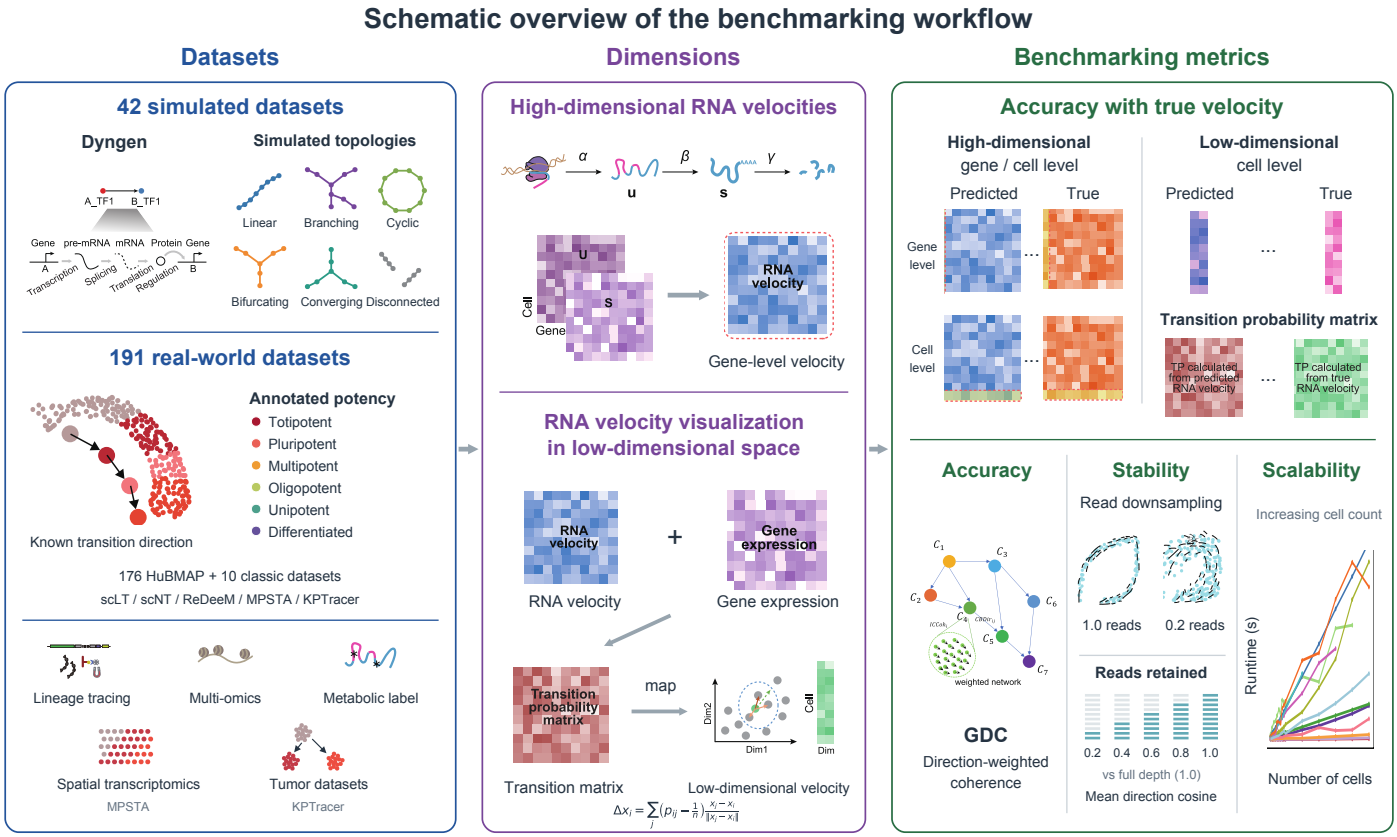


Fig. 2 | Schematic overview of the benchmarking workflow. The schematic depicts the 42 dyngen datasets used in the main simulation comparison, 176 HuBMAP datasets, ten classical datasets, and five task-specific inputs: scLT, scNT, ReDeeM, MPSTA and KPTracer. The original 30 cellDancer datasets are evaluated separately in the source-data summary (representative conditions in Extended Data Fig. 4c) and do not enter the main composite ranking. Counts in the schematic refer to dataset inputs, not independent studies or biological replicates. Spatial transcriptomics (MPSTA) and tumor data (KPTracer) are included alongside lineage tracing, multi-omics and metabolic labeling. The workflow covers high-dimensional RNA velocities, low-dimensional visualization and transition probability matrices. Accuracy is assessed using true velocities where available and direction-weighted coherence (GDC); stability compares predictions at 0.2, 0.4, 0.6 and 0.8× reads with full-depth predictions; scalability measures runtime as cell number increases. Stability assesses prediction consistency, not biological accuracy; the method-specific direction definitions, including the TopicVelo transition-projection proxy, are described in the Methods. Read-depth ratios are illustrated by ten short segments per stack; colored segments show the retained fraction and gray segments the removed fraction, with 1.0 as the full-depth reference. Flow fields, read segments and runtime curves shown here are schematic. Data sources are listed in Supplementary Tables 2 and 3.